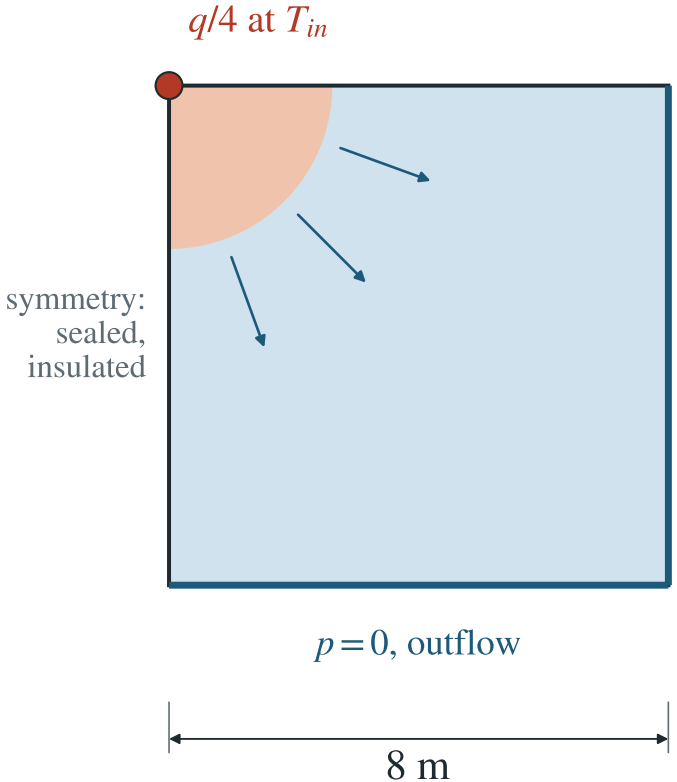
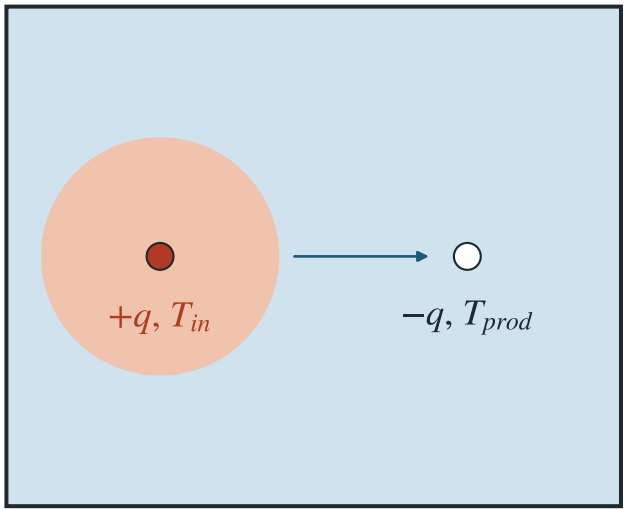
(a) radial injection, Avdonin



(b) sealed hot doublet

$$E - E_0 = \rho_f c_f q \int (T_{in} - T_{prod})\, dt$$



all sides: $q_n = 0$, $\partial T/\partial n = 0$

initially $T = T_0$, $p = 0$; $g = 0$; plan views